

Complementary traffic pattern analysis method using a FCD-based traffic information in online map

Presented by : Sanghoon JI, Kyunghee University

Authors :

Sanghoon JI, Kyunghee University

Jeong Chang Seong, University of West Georgia

Chul sue Hwang, Kyunghee University

Ana Stanescu, University of West Georgia



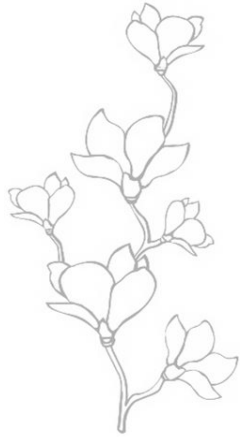
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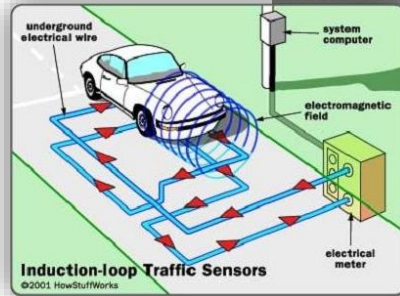
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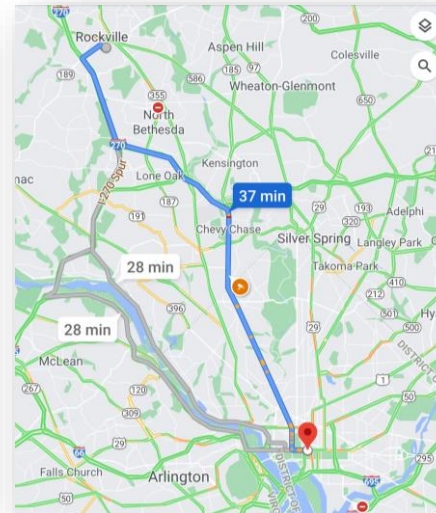
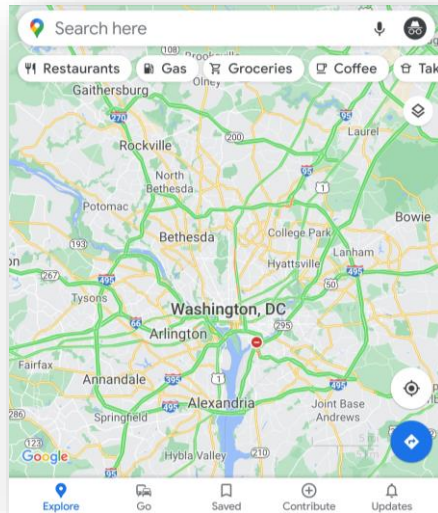
What traffic data are we using?

- ✓ Sensor-based data collection for traffic research



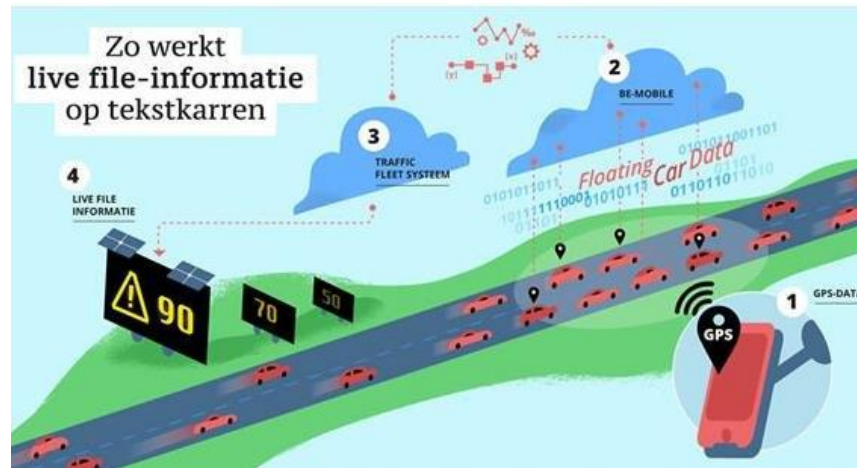
What traffic data are we using?

- ✓ Real-time Traffic information from Google Map



What is Floating Car Data ?

- ✓ Estimating traffic condition from crowd location data
 - Also called Crowdsourced traffic data



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Step 1. Comparison Analysis with conventional sensor-based traffic data

- ✓ How accurate Google Traffic information?
- ✓ What kind of sensor for traffic data collection?
 - Traffic Volume Monitoring Sensor
 - Vehicle Speed Monitoring Sensor

Step 2. Study on supplement methods for traffic pattern analysis

- ✓ What are limitations of sensor-based traffic data collection?
 - Sensors are only located in interstate highways
 - Unable to monitor another local roads
 - Sensors are located sparsely along the highway
 - Unable to analyze detailed traffic pattern of road sections between sensors
- ✓ Visualize the distribution of road sections for each traffic pattern
- ✓ Analyze road sections of county local roads that have not been monitored so far

Spatio-temporal data modeling of Google Traffic Information

✓ Data collection

- Snapping Google Maps image
- From January 31st to March 25th, 2021 (41 Days)
- Collected every 10 minutes

< URL used for map image collection >

`www.google.com/maps/@39.1269755,-77.1876252,12z/data=!5m1!1e1\`

Latitude

Longitude

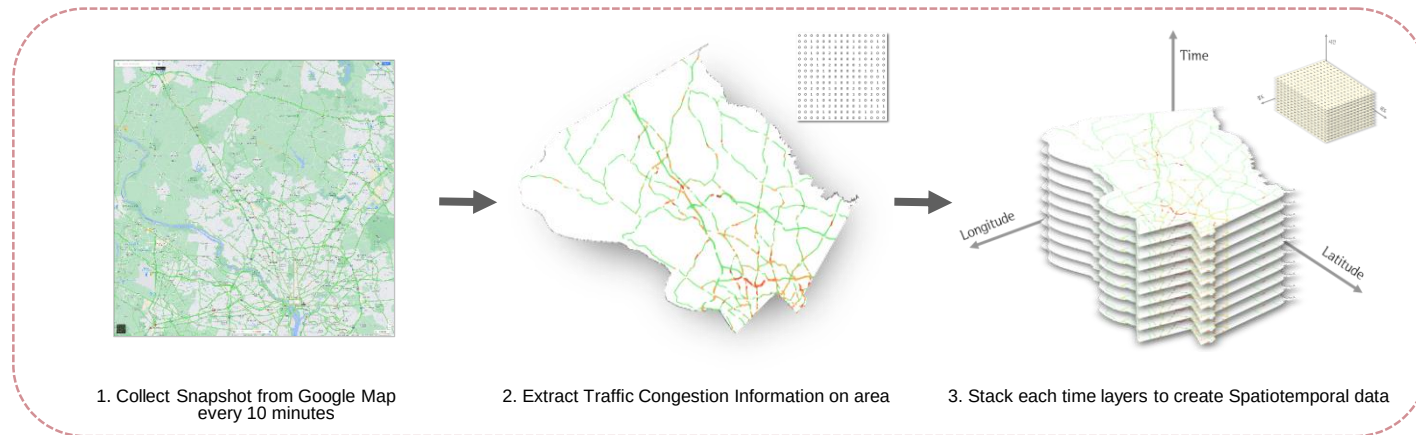
Scale

Traffic Layer

Spatio-temporal data modeling of Google Traffic Information

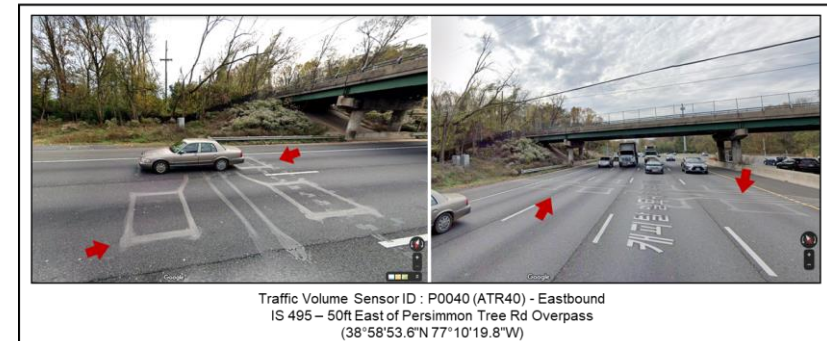
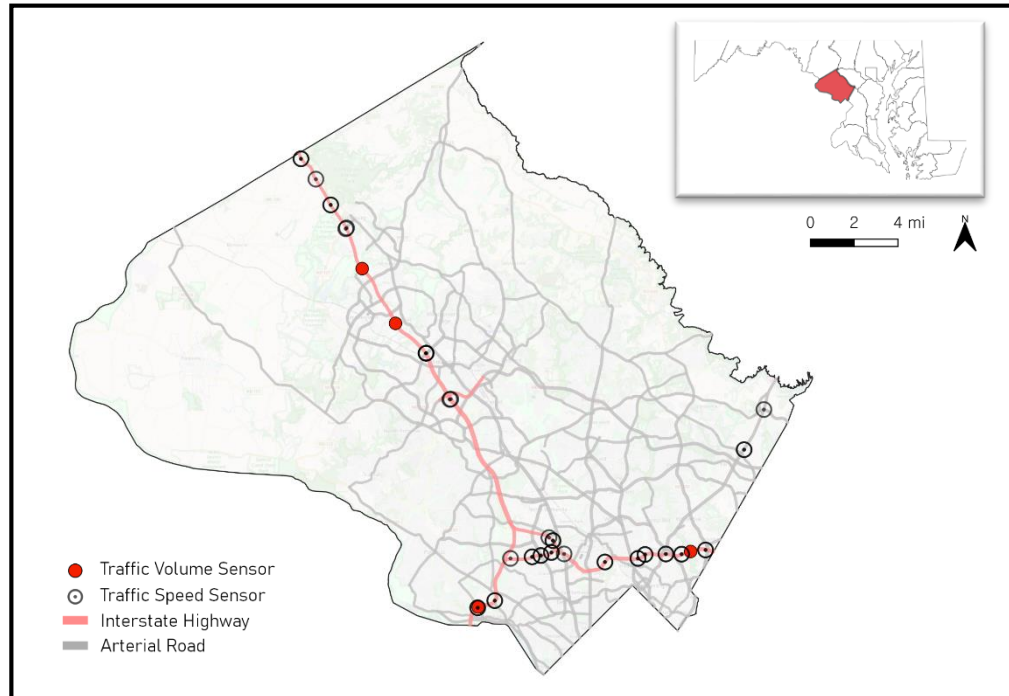
✓ Data modeling

- Processed with Python libraries – Numpy, OpenCV, Geopandas ...
- Stacked up into 3-dimentional spatio-temporal data
 - Numpy's multi-dimensional array structure
 - X axis : Longitude / Y axis : Latitude / Z axis : Time
- Replace each congestion indicate color into Congestion Index
 - Orange : 1 / Red : 2 / Dark Red : 4

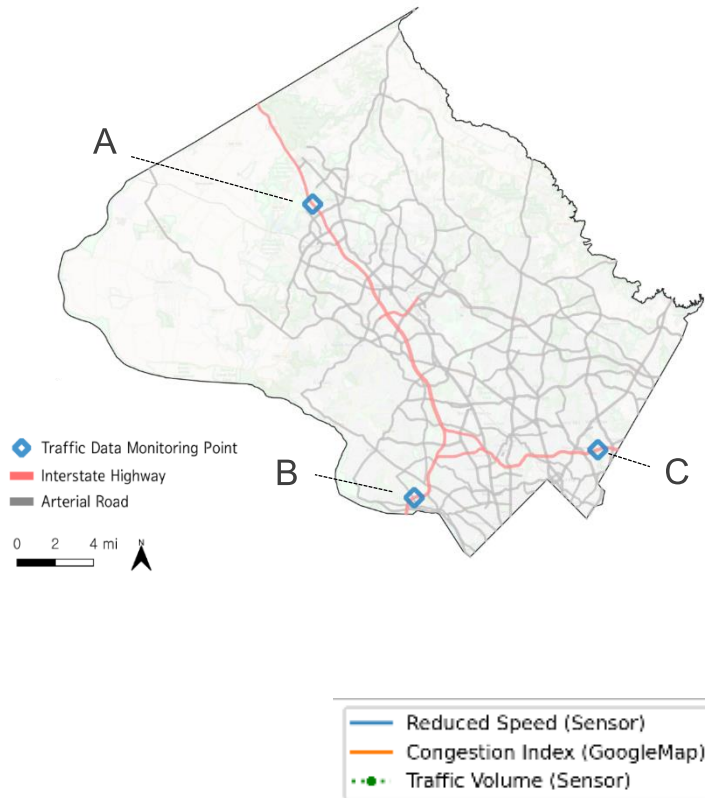


(1) Data Comparison

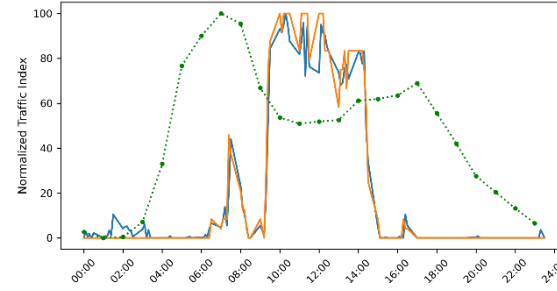
- ✓ Traffic Volume Sensor / Vehicle Speed Sensor / Google Traffic Information
- ✓ Verifying correlation and accuracy of Google Traffic Information
- ✓ From March 15th to March 17th, 2021 (3 days)



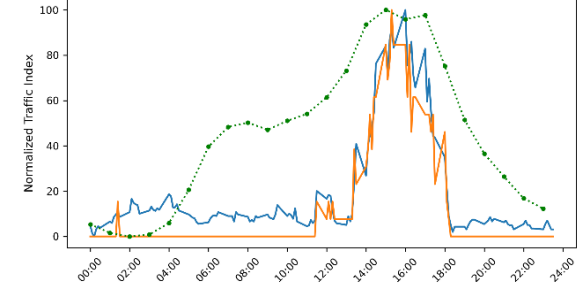
(2) Comparison Result



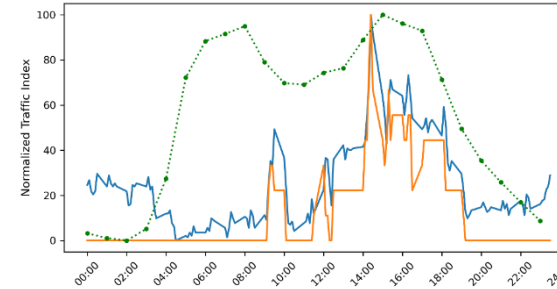
A - Southbound



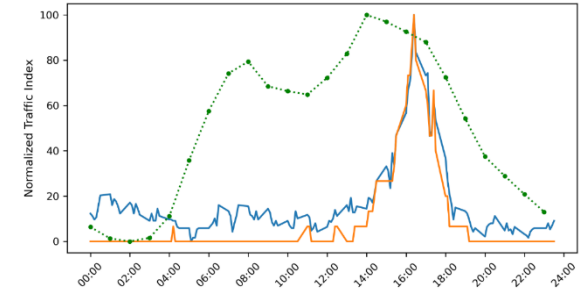
A - Northbound



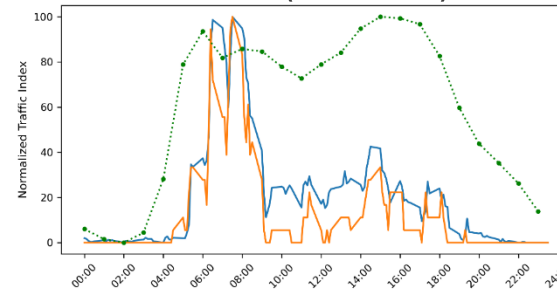
B - Westbound (Outer line)



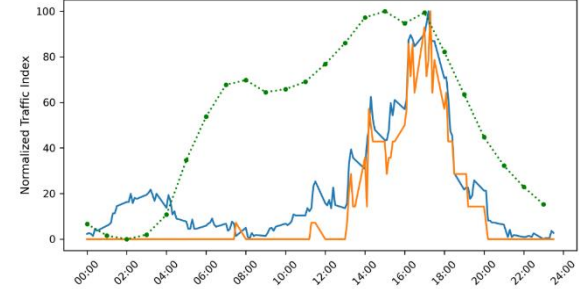
B - Eastbound (Inner Line)



C - Westbound (Outer Line)



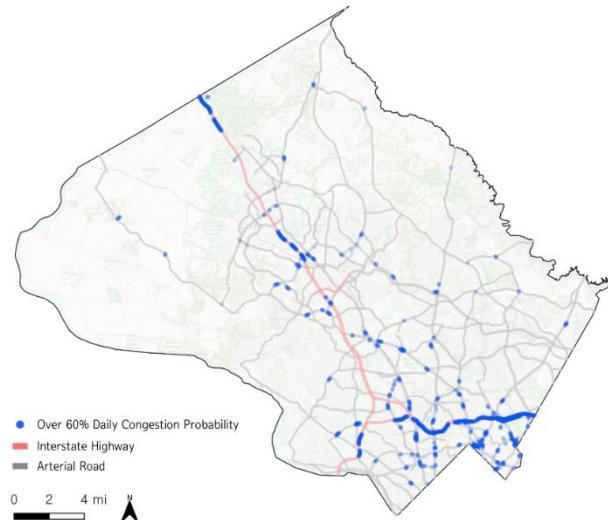
C - Eastbound (Inner Line)



(1) Expansion of monitoring road range

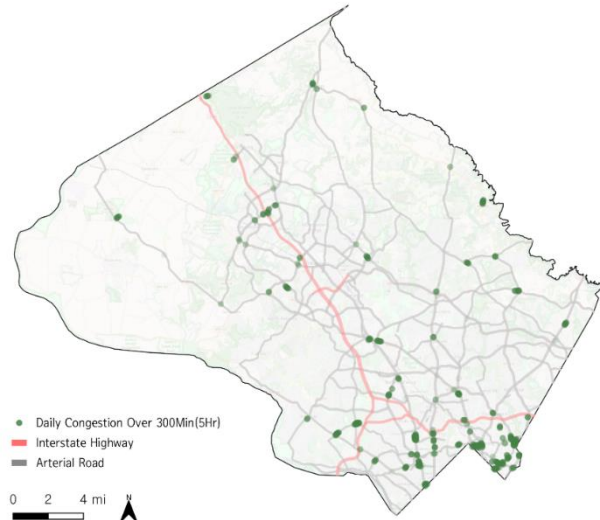
- ✓ Analyze the road sections outside of interstate highways

Road sections with high probability



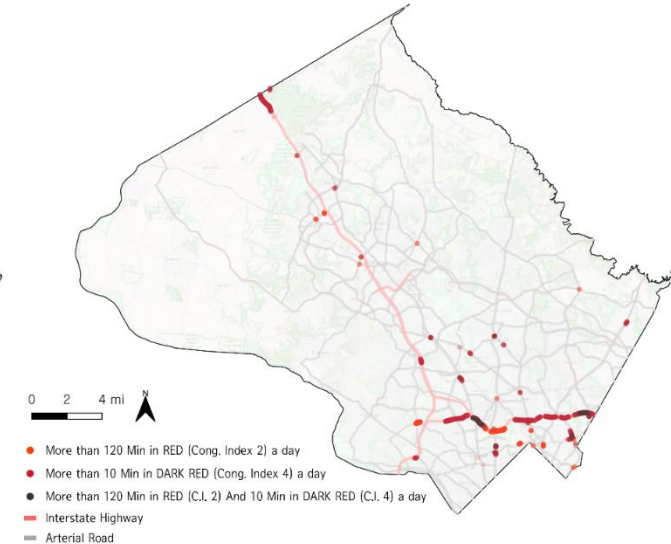
*** Probability :**
Days congestion occurred at a particular time / Total days monitored
 - Concentrated around Interstate 495 and urban areas in southern Montgomery county

Road sections with long-lasting congestion pattern



*** Daily averaged duration of congestion**
 - Concentrated in urban areas outside of highways
 - Long lasting congestion pattern, but not that severe

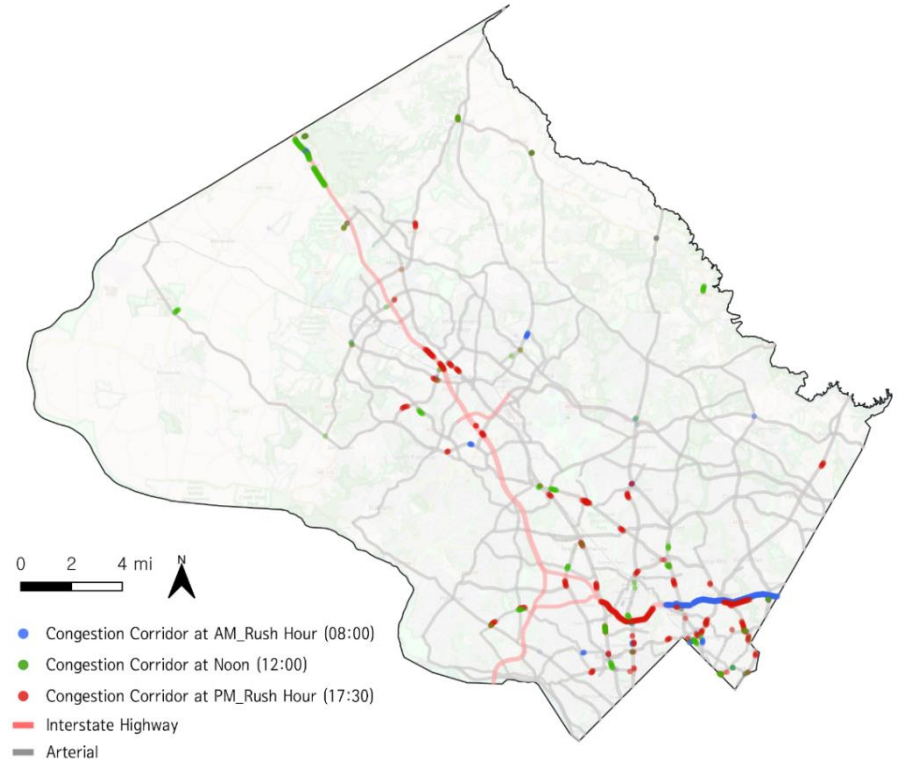
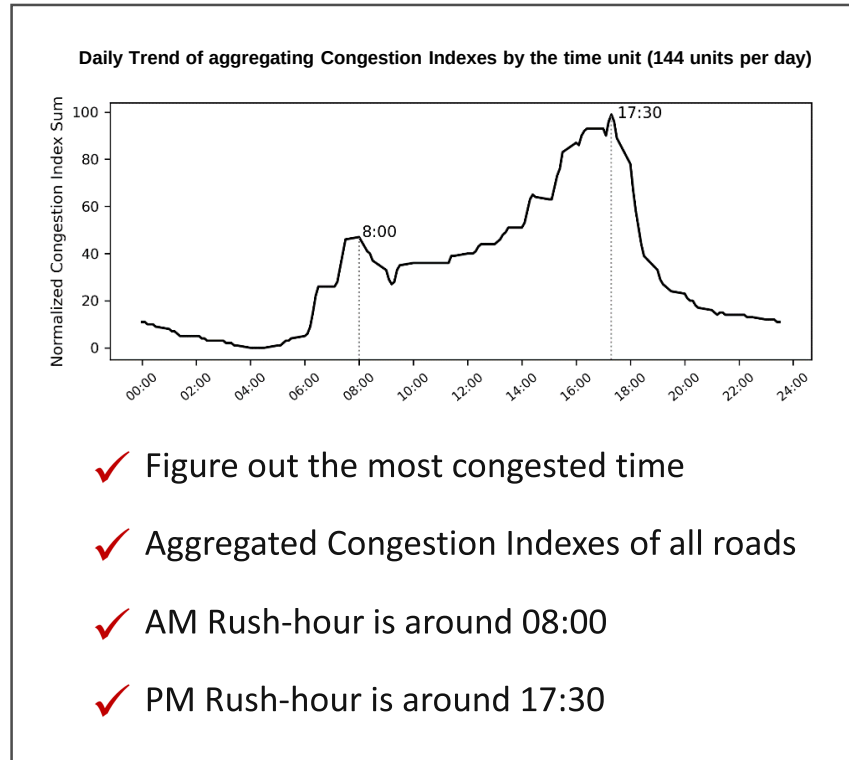
Road sections with severe congestion pattern



*** Daily averaged number of each Congestion Index**
 - Concentrated on Interstate 495 highways
 - Severe congestion pattern, but not that long-lasting

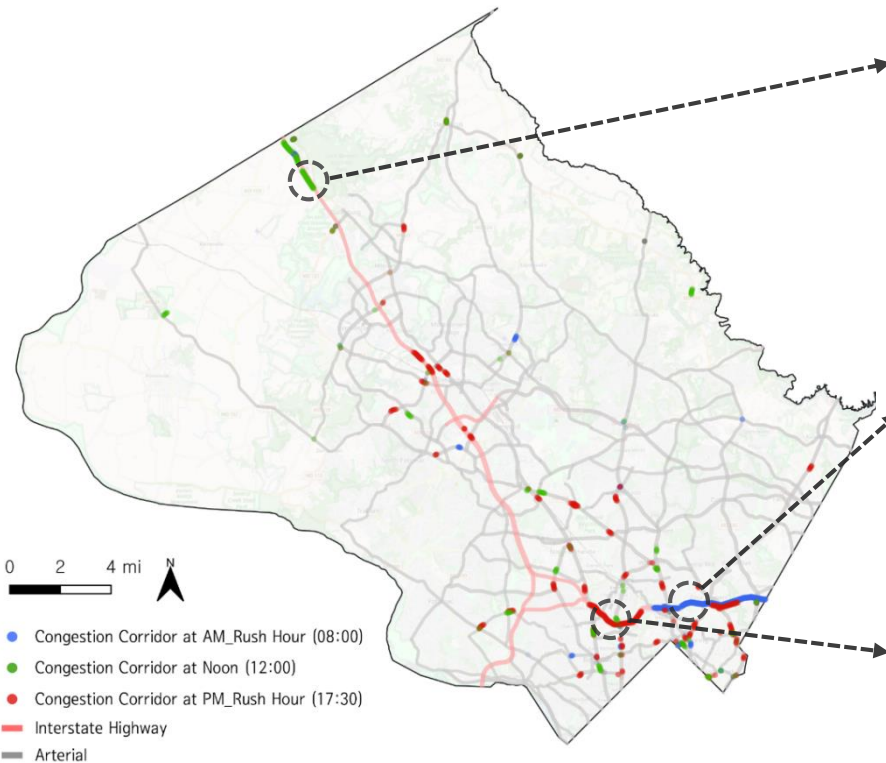
(2) Detailed congestion pattern analysis of road sections

✓ AM/PM Rush-hour congestion pattern analysis

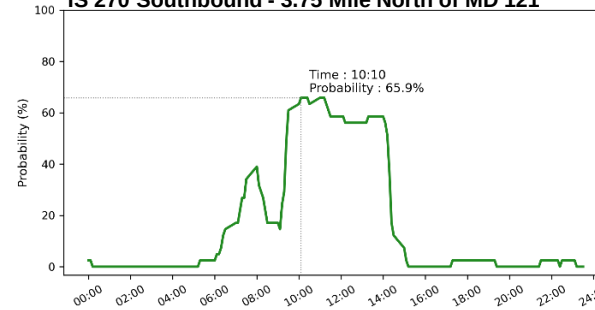


(2) Detailed congestion pattern analysis of road sections

- ✓ Daily congestion trend on the right plots
- ✓ Road points that couldn't be monitored



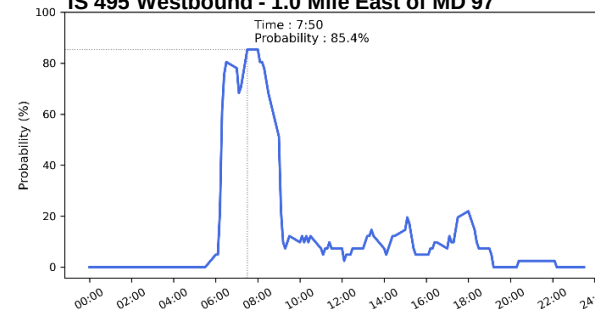
IS 270 Southbound - 3.75 Mile North of MD 121



Daily Congestion Index (C.I.) Statistics

Most Congested Time : 10 : 10
Probability : 65.9 %
C.I. Sum: 30.7
C.I. Mean : 1.3 (orange : 1 / red : 2 / dark red : 4)
C.I. Duration : 231 min
Cong. Time Zone : 09:50 – 14:20
Number of C.I. 1,2,4 :
C.I. 1 (Orange) : 158 min
C.I. 2 (Red) : 72 min
C.I. 4 (Dark Red) : 1 min

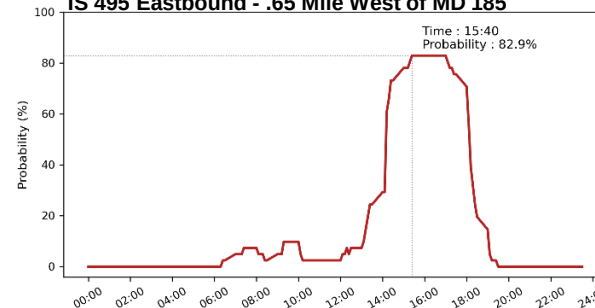
IS 495 Westbound - 1.0 Mile East of MD 97



Daily Congestion Index (C.I.) Statistics

Most Congested Time : 07:50 – 08:00
Probability : 85.4 %
C.I. Sum: 25.4
C.I. Mean : 1.4 (orange : 1 / red : 2 / dark red : 4)
C.I. Duration : 187 min
Cong. Time Zone : 06:30 – 09:00
Number of C.I. 1,2,4 :
C.I. 1 (Orange) : 139 min
C.I. 2 (Red) : 38 min
C.I. 4 (Dark Red) : 10 min

IS 495 Eastbound - .65 Mile West of MD 185



Daily Congestion Index (C.I.) Statistics

Most Congested Time : 15:40 – 17:00
Probability : 82.9 %
C.I. Sum: 33.2
C.I. Mean : 1.4 (orange : 1 / red : 2 / dark red : 4)
C.I. Duration : 233 min
Cong. Time Zone : 14:20 – 18:10
Number of C.I. 1,2,4 :
C.I. 1 (Orange) : 143 min
C.I. 2 (Red) : 86 min
C.I. 4 (Dark Red) : 5 min

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- ✓ Research for utilizing Floating Car Data method (Google Maps traffic data)
- ✓ Googlemap stacked Spatio-temporal data model for traffic pattern analysis
- ✓ Google traffic information can complement the limitations of the sensor-based traffic data
 - ✓ Able to analyze on the road sections that sensors are not installed
- ✓ Google traffic information enables to conduct detailed traffic pattern analysis on a wide range of areas.

2021 AAG Annual meeting

Complementary traffic pattern analysis method using a FCD-based traffic information in online map

Presented by : Sanghoon Ji

jish@khu.ac.kr

Geography, GIS

Department of Geography

Kyunghee University, Seoul, Korea

Thank You



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